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The blind Bach-maker

Keith Frankish

From Bacteria to Bach and Back: The Evolution of Minds. By Daniel C. Dennett. W. W. Norton / Allen Lane, 2017. 496 pp.

Our neurons are cousins of yeast cells and descendants of even simpler microbes, yet they are organised into structures capable of astonishing feats of creativity. How did we get from bacteria to Bach, from fungus to fugues? Daniel Dennett tells the story, revising and extending 50 years' work on the topic.

The story is one of Darwinian natural selection — of design emerging gradually from blind processes of replication and differential selection — but it also involves some bold switches of perspective ("strange inversions of reasoning"), designed to loosen the pull of "Cartesian gravity" — our introspective conception of the mind as mysterious and non-physical.

A key slogan is "competence without comprehension". Just as computers can perform complex calculations without understanding arithmetic, so creatures can display finely tuned behaviours without understanding why they do so. The rationales for their behaviours (diverting a predator, say, or tempting a mate) are "free-floating" — implicit in the creatures' design but not represented in their minds. Competence without comprehension is the default in nature, Dennett argues, even among higher animals.

How then did human intelligence arise? We do not have a special faculty of comprehension. (Dennett says that comprehension is *composed of* competences.) Rather, our minds have been enhanced by a process of cultural evolution, operating on *memes*. Memes are copyable behaviours — words (ways of vocalising) being paradigmatic examples. Initially, memes spread in human populations like viruses, selected simply for their infectiousness. Some were useful, however, and our brains adapted to foster them, genetic and memetic evolution working together. Words and other memes gifted humans with powerful new competences — for communication, explicit representation, reflection, self-interrogation and self-monitoring. In this way, human culture "de-Darwinized" itself. (Dennett nicely illustrates how Darwinian processes have various dimensions and degrees.) Memetic evolution provided "thinking tools" — a bit like smartphone apps — which transformed humans into comprehending, intelligent designers, triggering an explosion of civilisation and technology.

Dennett sees human consciousness, too, as a product of both genetics and memetics. The need to communicate — and sometimes withhold — our thoughts fostered schematic introspective representations — an "edited digest" of our cognitive processes, which serves as the brain's own "user interface". The mental items that populate our consciousness are the fictional objects of these representations.

The book concludes with a look forwards. Dennett doubts that superintelligent AI will dominate us but worries that we may overestimate the intelligence of our artefacts and become over-reliant on them, and that the cultural structures on which human comprehension depends may erode.

This summary only hints at the range and richness of the book. Dennett provides illuminating explanations of the concepts and principles he employs and cites fascinating experimental work. Many of the book's theoretical claims are controversial — especially those about memetics and consciousness — but Dennett is well aware of the objections and addresses them with patience. Some readers will be more persuaded than others, but Dennett has an excellent record of predicting developments in cognitive science, and it would be rash to bet that he is far off the right track. Persuaded or not, readers will find their minds enriched with many new and powerful thinking tools.